Final report: 20 page report; decent literature review
(key papers, intro, why the interest,
methods + results, discussion, conclusion)

Geometrical alg to match surface of two solids
(w/ diff. lattice constants & periodicities).

Electrocatalysis Cont'd

ORR: 1) $O_2(g) + H^+ + e^- + * \longleftrightarrow OOH^*$

2) $OOH^* + H^+ + e^- \longleftrightarrow O^* + H_2O(g)$

3) $O^* + H^+ + e^- \longrightarrow OH^*$

4) $OH^* + H^+ + e^- \longleftrightarrow H_2O(g) + *$

Find ΔG_1, ΔG_2, ΔG_3, ΔG_4 in terms of ΔE_o

Use "limiting legs" analysis, combined w/ Sabatier Analysis (assume idealized surface energies)

Step 1 is RLS: (potential)

$$r = r_1 = k_1(U)\, P_{O_2}\, C_{H^+}\, \theta_*^2 - k_{-1}(U)\, \theta_{OOH}$$

$$= k_1 P_{O_2} C_{H^+} \theta_*^2 \left[1 - \underbrace{\frac{k_{-1}}{k_1} \frac{\theta_{OOH}}{P_{O_2} C_{H^+} \theta_*^2}}_{\gamma_1} \right]$$

Assume maximum rates: irreversibilities & idealized θ's = 1

Similarly:

$$r_1^{max} = k_1 P_{O_2} C_{H^+}$$

$$r_2^{max} = k_2 C_{H^+}$$

$$r_3^{max} = k_3 C_{H^+}$$

$$r_4^{max} = k_4 C_{H^+}$$

Example for k_1: voltage-dependent
(use BEP relation)

$$\rightarrow k_1 = \frac{k_{1o}}{c_{H^+}^\circ P^\circ} \exp\left(-\alpha \left(\frac{\Delta G_1^\circ + eU}{k_B T} \right) \right)$$

(ΔG_1°: voltage dependent; usually $\frac{1}{2} < \alpha < 1$; Pre-factor: Double layer)

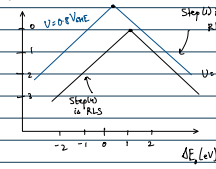
assume $\alpha = 1$ for now

compared in terms of ΔE_o

$$-k_B T \ln\left(\frac{r_1}{k_{1o}}\right) = \Delta G_1^\circ + e V_{SHE} - k_B T \ln\left(\frac{P_{O_2} C_{H^+}}{P^\circ C_{H^+}^\circ}\right)$$

k_{1o} is a pre-exponential factor for H^+ transfer through double layer. Use TST or fit to experiment

Plot activity vs. ΔE_o for given V_{SHE}:



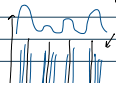
+ U affects location of max a little
+ Rate increases as U decreases (but cell efficiency decreases)

You want the potential difference b/w the anode & cathode to be the highest (but the rate is slow for when operating at 1.23V)

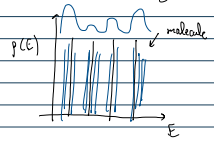
D-band Analysis

KS equations; KS wave functions → eigenvalues (ε_i) → app to energies of e^- orbitals.

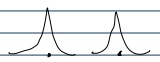
Density of states → electronic properties



Density of states → electronic properties



Norskov d-band theory: tight binding (molecular-orbital theory)



$\Psi_{AB} = C_A \Psi_A + C_B \Psi_B$

↳ Can have more orbitals than particles also be involved.

$H_{AB} = -\frac{1}{2}\nabla^2 + v_A + v_B$

1 e^- Hamiltonian (approximate) → Matrix form & the set det. equal to zero → Formal way of understanding molecular orbital theory.

Newns-Anderson theory

↳ esp. for metals (using density of states instead of metal orbitals)

So, not completely ignoring s & p bands; d-band acts like a "molecule"

↳ just prioritizing d-band?

↳ expression is same as molecular orbital theory exp. (ΔE)

$\int, V_{ad}^2$ } before binding parameters

E_0 → estimate (interaction w/ s-p band)

E_1 → d-band center (i.e. independent of ads.)

E_a →

$f=1$ (full d-band)

↳ post-tran

meta